Structural oliGARCH Dynamics in a Human–AI Knowledge Economy

Strategic AI Competition, Endogenous Volatility, and Innovation Timing in a General Equilibrium Framework

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Abstract

We develop a structural general equilibrium model of a knowledge economy in which N AI firms compete oligopolistically in the Cournot tradition, and households invest in both consumption and human capital. Knowledge production follows stochastic productivity whose volatility is governed by a *structural oliGARCH mechanism*: a GARCH-type recursion augmented with a strategic-dispersion term that directly captures how the cross-sectional variance of AI firm outputs amplifies aggregate innovation uncertainty. We derive (i) closed-form Cournot–Nash equilibria in the knowledge market, (ii) stability conditions for the variance process via Lyapunov methods, (iii) identification and maximum-likelihood consistency results, (iv) an operator-theoretic characterisation of the volatility recursion, and (v) a spectral-gap formula—via the Witten Laplacian—that determines the expected inter-arrival time of innovation bursts. An illustrative calibration exercise matches the model's implied volatility clustering and knowledge price dynamics to stylised facts from the AI productivity literature. Market concentration (N) emerges as a key bifurcation parameter: reducing AI oligopoly power lowers both the equilibrium price of knowledge and the risk of volatility blow-up.

The paper ends with "The End"

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1 Introduction

The rapid expansion of large-scale artificial intelligence has transformed the nature of knowledge production. Where knowledge was once generated almost exclusively by human researchers, it is increasingly co-produced—or entirely produced—by AI systems whose output scales with computational resources rather than with human population. This shift poses challenges both for economic theory and for empirical macroeconomics. Standard endogenous-growth models (Romer, 1990; Aghion and Howitt, 1992) take knowledge production to be a smooth function of human effort; they offer little guidance on how AI's stochastic productivity, strategic market structure, and volatility clustering interact to shape macroeconomic dynamics.

This paper proposes a structural model that integrates three bodies of literature: (i) *endogenous growth theory*, in which knowledge is the engine of long-run productivity; (ii) *oligopoly theory*, which governs the strategic interactions of a small number of AI producers; and (iii) *GARCH econometrics*, which models the heteroskedastic and clustered nature of productivity shocks. The portmanteau term *oliGARCH* is meant to capture precisely this synthesis: an oligopolistic knowledge market whose aggregate volatility follows a generalised autoregressive conditionally heteroskedastic process augmented with a term that measures strategic cross-sectional dispersion.

1.1 Main contributions

The paper makes four specific contributions.

(1) Cournot AI oligopoly. We model the knowledge sector as N AI firms competing à la Cournot in a market for knowledge services. Each firm chooses how much knowledge to produce, taking rivals' outputs as given. We derive closed-form symmetric equilibria, show uniqueness via the contraction-mapping argument of Vives (2001), and characterise how market concentration interacts with productivity shocks.

(2) Structural oliGARCH. The aggregate volatility of knowledge production obeys a new recursion that nests standard GARCH$(1, 1)$ as a special case but adds a strategic-dispersion term measuring the cross-sectional variance of firm-level outputs. We prove covariance stationarity using a Lyapunov-function argument and derive an exact expression for the unconditional variance.

(3) Operator theory and spectral analysis. We represent the volatility recursion as a nonlinear operator on a Banach space and characterise its fixed point via the Banach Fixed-Point Theorem. Applying the Witten Laplacian formalism from mathematical physics (Witten, 1982; Holley and Stroock, 1987), we derive an explicit spectral-gap formula governing the expected inter-arrival time of innovation bursts.

(4) Calibration. We calibrate the model to stylised moments from AI compute scaling and R&D productivity data, show that the stability condition is satisfied at empirically plausible parameter values, and compute the implied innovation-burst frequency.

1.2 Organisation

Section 2 reviews the literature. Section 3 describes the economic environment. Section 4 solves the household problem. Section 5 derives the Cournot–Nash equilibrium in the knowledge market. Section 6 defines and characterises the competitive equilibrium. Section 7 develops the oliGARCH mechanism. Section 8 establishes identification and estimation results. Section 9 provides the operator-theoretic treatment. Section 10 derives the spectral gap. Section 11 presents the calibration. Section 12 discusses policy implications and extensions. Section 13 concludes.

2 Related Literature

2.1 AI and economic growth

The macroeconomic consequences of AI have attracted growing theoretical and empirical attention. Acemoglu and Restrepo (2018) model robots as a factor of production that substitutes for human labour, showing that automation need not raise aggregate wages or welfare. Acemoglu and Restrepo (2019) extend this analysis to a task-based framework, demonstrating that AI's displacement effect can outweigh its productivity effect at current rates of diffusion. Aghion et al. (2019) embed AI in a Schumpeterian growth model to argue that automation could sustain long-run growth, but only if human complementarity in idea production is preserved. Brynjolfsson et al. (2018) emphasise the near-term costs of learning and reorganisation that may delay the aggregate productivity gains from AI. Our model complements this literature by focusing on the market structure of AI knowledge producers and the stochastic volatility of their output, dimensions that have received less formal attention.

2.2 Endogenous growth

Our framework builds on Romer (1990), who showed that knowledge exhibits increasing returns and acts as the engine of sustained per capita growth. Aghion and Howitt (1992) formalise quality ladders in which innovation involves creative destruction. Jones (1995) revises Romer by introducing diminishing returns to research, generating semi-endogenous growth. In our model, knowledge is produced by AI firms rather than human researchers, but the long-run mechanics are analogous: the equilibrium knowledge stock and its growth rate are endogenously determined by market structure and productivity.

2.3 Oligopoly and innovation

The relationship between market concentration and innovation is a venerable question in industrial organisation. Arrow (1962) argued that competition spurs innovation, while Schumpeter (1942) emphasised the role of market power in financing R&D. Aghion et al. (2005) reconcile these views with an inverted-U relationship between competition and innovation intensity. Our Cournot knowledge market follows the classic treatment in Vives (2001), adapted to the AI production environment.

2.4 GARCH and stochastic volatility

The GARCH model of Engle (1982) and its generalisation by Bollerslev (1986) have become the workhorses for modelling time-varying volatility in financial markets. Nelson (1991) proposes the EGARCH model, allowing asymmetric responses to positive and negative shocks. The structural GARCH approach, relating volatility to economic fundamentals, is developed by Engle (2001). Our oliGARCH mechanism extends structural GARCH by incorporating a knowledge-market feedback channel and a strategic-dispersion term with explicit microeconomic foundations.

2.5 Spectral methods in economics

The use of spectral theory to analyse economic dynamics traces back to Brock (1972). Applications of the Witten Laplacian—originally developed in the context of supersymmetric quantum mechanics (Witten, 1982)—to diffusion processes and metastability are developed by Holley and Stroock (1987) and Bovier et al. (2004). We adapt these tools to quantify the timing of innovation cascades in our knowledge economy.

3 Economic Environment

Time is discrete and indexed by $t \in \{0, 1, 2, \ldots\}$. The economy contains three types of agents: a unit mass of identical households, a final-goods firm, and $N \geq 1$ AI knowledge-producing firms.

Commodities

There are three goods each period: a final consumption good (numéraire), a knowledge service, and labour. The knowledge service is produced by AI firms and consumed by households (directly) and by the final-goods firm (as a productive input).

Notation

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space. For any random variable X_t, let $\mathbb{E}_t[\,\cdot\,]$ denote expectation conditional on the information set $\mathcal{F}_t = \sigma\{X_s : s \leq t\}$.

4 Households

4.1 Preferences and Budget

Households derive utility from consumption C_t and the stock of knowledge services K_t^d they purchase, and discount the future at rate $\beta \in (0, 1)$. Human capital H_t enters the household's labour supply and its investment decision.

The household solves:

$$\max_{\{C_t,\, K_t^d,\, I_{H,t}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}}{1-\sigma} + \chi \, \frac{(K_t^d)^{1-\rho}}{1-\rho} \right] \tag{1}$$

subject to the period budget constraint

$$C_t + p_t K_t^d + I_{H,t} = w_t H_t + \Pi_t, \tag{2}$$

the human-capital law of motion

$$H_{t+1} = (1 - \delta_H)\, H_t + \phi\, I_{H,t}, \tag{3}$$

and non-negativity constraints $C_t, K_t^d, I_{H,t} \geq 0$. Here $\sigma > 0$ is the inverse elasticity of intertemporal substitution, $\rho > 0$ governs the demand elasticity for knowledge, $\chi > 0$ is the weight on knowledge utility, $\delta_H \in (0, 1)$ is the depreciation rate of human capital, $\phi > 0$ is the learning efficiency parameter, w_t is the wage, p_t is the price of knowledge, and Π_t denotes dividends received from AI firms (households own the firms in equal shares).

4.2 Optimality Conditions

The Lagrangian for the household's problem, with λ_t and μ_t as multipliers on (2) and (3) respectively, yields the following first-order conditions:

$$C_t^{-\sigma} = \lambda_t \tag{4}$$
$$\chi (K_t^d)^{-\rho} = \lambda_t p_t \tag{5}$$
$$\mu_t = \lambda_t \tag{6}$$
$$\mu_t = \beta \mathbb{E}_t [\lambda_{t+1} w_{t+1} + (1 - \delta_H)\, \mu_{t+1}] \tag{7}$$

Proposition 4.1 (Household Optimality)**.** *The optimal knowledge demand satisfies*

$$K_t^d = \left(\frac{\chi}{C_t^{-\sigma}\, p_t} \right)^{1/\rho}, \tag{8}$$

and the Euler equation for human capital is

$$1 = \beta\, \mathbb{E}_t \left[C_{t+1}^{-\sigma} / C_t^{-\sigma}\ (w_{t+1}\phi + (1 - \delta_H)) \right]. \tag{9}$$

Proof. Dividing (5) by (4) gives $\chi (K_t^d)^{-\rho} = C_t^{-\sigma} p_t$, which rearranges to (8). Substituting (6) and (4) into (7) and leading by one period yields (9). □

Remark 4.1. *With $\rho = 1$ (log utility over knowledge) and ignoring the human-capital margin, demand reduces to $K_t^d = \chi C_t^{\sigma} / p_t$, proportional to the ratio of real wealth to the knowledge price—recovering the spirit of the baseline model in the original paper.*

5 AI Knowledge Production and Cournot Competition

5.1 Technology

There are N AI firms, indexed $i \in \{1, \ldots, N\}$. Firm i allocates computational resources $l_{i,t} > 0$ to produce knowledge:

$$k_{i,t} = \theta_{i,t}\, l_{i,t}, \tag{10}$$

where $\theta_{i,t}$ is firm i's stochastic productivity. We decompose productivity into an aggregate and an idiosyncratic component:

$$\theta_{i,t} = \theta_t + \xi_{i,t}, \qquad \xi_{i,t} \overset{\text{iid}}{\sim} \mathcal{N}(0,\, \sigma_\xi^2), \tag{11}$$

where $\theta_t = \mu + \sigma_t \varepsilon_t$ with $\varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$ is the common aggregate component and $\xi_{i,t}$ is a firm-specific innovation shock, independent across firms and time.

Total knowledge supplied is

$$K_t^s = \sum_{i=1}^{N} k_{i,t}. \tag{12}$$

5.2 Market Demand and Inverse Demand

The final-goods firm (described below) and households together generate an aggregate knowledge demand schedule. We derive the following linear inverse demand:

Assumption 5.1 (Linear Inverse Demand)**.** *Aggregate inverse demand for knowledge takes the form*

$$p(K_t;\, W_t) = \frac{W_t}{\bar{K}} - b\, K_t, \tag{13}$$

where $W_t = w_t H_t + \Pi_t$ denotes household wealth, $\bar{K} > 0$ is a normalisation constant, and $b > 0$ is the demand slope. The intercept $a_t \equiv W_t / \bar{K}$ is stochastic and wealth-driven.

Linear demand arises as the first-order approximation to any smooth, downward-sloping demand curve around the steady state. It is standard in the Cournot oligopoly literature (Vives, 2001).

5.3 Firms' Costs

Each firm i incurs a convex cost of computational resource allocation:

$$c_i(k_{i,t},\, \theta_{i,t}) = \frac{c}{2} \frac{k_{i,t}^2}{\theta_{i,t}}, \tag{14}$$

where $c > 0$ is a cost parameter. This quadratic form captures diminishing returns to scale in compute: each additional unit of knowledge output requires proportionally more compute. When $\theta_{i,t}$ is high (high productivity), the marginal cost is lower.

5.4 Cournot–Nash Equilibrium

Firm i maximises profit taking rivals' outputs $K_{-i,t} = \sum_{j \neq i} k_{j,t}$ as given:

$$\max_{k_{i,t} \geq 0} \pi_{i,t} = p(K_t;\, W_t)\, k_{i,t} - \frac{c}{2} \frac{k_{i,t}^2}{\theta_{i,t}}. \tag{15}$$

The first-order condition is

$$a_t - b\,(K_{-i,t} + k_{i,t}) - b\, k_{i,t} - \frac{c\, k_{i,t}}{\theta_{i,t}} = 0, \tag{16}$$

yielding firm i's best-response function:

$$k_{i,t}^{\mathrm{BR}}(K_{-i,t}) = \frac{\theta_{i,t}(a_t - b\, K_{-i,t})}{2b\, \theta_{i,t} + c}. \tag{17}$$

Theorem 5.1 (Existence and Uniqueness of Cournot–Nash Equilibrium)**.** *Under Assumption 5.1 and $a_t, \theta_{i,t}, c, b > 0$, there exists a unique Nash equilibrium $\mathbf{k}_t^* = (k_{1,t}^*, \ldots, k_{N,t}^*)$ in pure strategies. The symmetric equilibrium (when $\theta_{i,t} = \theta_t$ for all i) is:*

$$k_t^* = \frac{a_t\, \theta_t}{(N+1)\, b\, \theta_t + c}, \tag{18}$$

$$K_t^* = \frac{N\, a_t\, \theta_t}{(N+1)\, b\, \theta_t + c}, \tag{19}$$

$$p_t^* = \frac{a_t\, (b\, \theta_t + c)}{(N+1)\, b\, \theta_t + c}. \tag{20}$$

Proof. Existence. For each i, the profit function $\pi_{i,t}$ is strictly concave in $k_{i,t}$ (since $-2b - c/\theta_{i,t} < 0$) and the strategy space $[0, \bar{k}]$ is compact. By Brouwer's fixed-point theorem applied to the best-response map, a pure-strategy Nash equilibrium exists.

Uniqueness. The best-response map $\mathbf{k} \mapsto \mathbf{k}^{\mathrm{BR}}(\mathbf{k})$ is a contraction. Indeed, for any two strategy profiles $\mathbf{k}$ and $\tilde{\mathbf{k}}$,

$$\left| \frac{\partial k_{i,t}^{\mathrm{BR}}}{\partial k_{j,t}} \right| = \frac{b\, \theta_{i,t}}{2b\, \theta_{i,t} + c} < \frac{1}{2} \quad \forall\, j \neq i,$$

so the aggregate contraction modulus satisfies $\sum_{j \neq i} |\partial k_i^{\mathrm{BR}} / \partial k_j| < N/(N+1) < 1$. By the Vives (2001) uniqueness theorem for supermodular games, the equilibrium is unique.

Symmetric equilibrium. Setting $\theta_{i,t} = \theta_t$ and imposing $k_{i,t} = k_t^*$ in (17):

$$k_t^* = \frac{\theta_t(a_t - b(N-1)k_t^*)}{2b\, \theta_t + c}.$$

Solving: $(2b\, \theta_t + c)\, k_t^* = \theta_t\, a_t - b(N-1)\theta_t\, k_t^*$, so $[(N+1)b\, \theta_t + c]\, k_t^* = a_t\, \theta_t$, yielding (18). Equations (19) and (20) follow by $K_t^* = N k_t^*$ and substituting into (13). □

Corollary 5.2 (Market Power and Knowledge Price)**.** *$\partial p_t^* / \partial N < 0$ and $\partial K_t^* / \partial N > 0$: more AI competitors reduce the price of knowledge and expand its supply. As $N \to \infty$, $p_t^* \to c\, a_t / (b\, \theta_t) \cdot b/(b) = c/(b\, \theta_t + c/(N \to \infty))$; in the limit, $p_t^* \to 0$ and $K_t^* \to a_t / b$ (competitive outcome).*

Proof. Differentiating (20):

$$\frac{\partial p_t^*}{\partial N} = \frac{-a_t\, b\, \theta_t (b\, \theta_t + c)}{[(N+1)b\, \theta_t + c]^2} < 0.$$

The sign of $\partial K_t^* / \partial N$ follows symmetrically from (19). The competitive limit is obtained by dividing numerator and denominator of (19) by N and letting $N \to \infty$. □

6 Final Goods and General Equilibrium

6.1 Final-Goods Firm

A representative final-goods firm operates a constant-returns-to-scale technology:

$$Y_t = Z_t\, (K_t^s)^{\phi}\, H_t^{1-\phi}, \qquad \phi \in (0, 1), \tag{21}$$

where $Z_t > 0$ is exogenous TFP and H_t is aggregate human capital. Competitive factor markets yield:

$$p_t^Y = \phi\, Z_t\, (K_t^s)^{\phi - 1}\, H_t^{1-\phi} \quad \text{(knowledge rental rate)}, \tag{22}$$

$$w_t = (1 - \phi)\, Z_t\, (K_t^s)^{\phi}\, H_t^{-\phi} \quad \text{(wage)}. \tag{23}$$

Remark 6.1. *The knowledge price p_t faced by households is a composite of p_t^Y (the productive value) and a direct utility value. In equilibrium, arbitrage ensures $p_t = p_t^Y$ up to a taste parameter.*

6.2 Equilibrium Definition

Definition 6.1 (Competitive Equilibrium)**.** *A competitive equilibrium is a sequence of prices $\{p_t, w_t, r_t\}_{t=0}^{\infty}$ and allocations $\{C_t, K_t^d, I_{H,t}, H_t, k_{i,t}, K_t^s, Y_t\}_{t=0}^{\infty}$ such that:*

1. *Households maximise (1) subject to (2)–(3), taking prices as given.*
2. *Each AI firm i plays the Cournot–Nash equilibrium (18).*
3. *The final-goods firm maximises profits, yielding (22)–(23).*
4. *Markets clear: $K_t^d = K_t^s$ (knowledge), $Y_t = C_t + I_{H,t}$ (goods), and H_t clears the labour market.*

The equilibrium knowledge price is pinned down by market clearing:

$$p_t = \frac{a_t\,(b\,\theta_t + c)}{(N+1)\,b\,\theta_t + c}, \tag{24}$$

where $a_t = W_t/\bar{K}$ is endogenously determined by wages and dividends.

7 The Structural oliGARCH Mechanism

7.1 Motivation: Why Oligopolistic Structure Amplifies Volatility

In the Cournot equilibrium, firm i's output responds to both aggregate and idiosyncratic shocks. When one firm experiences a positive productivity shock, its best-response shifts outward, which induces rivals to contract (strategic substitutes). This *strategic interaction* generates cross-sectional dispersion in outputs even in a symmetric setting whenever $\xi_{i,t} \neq 0$. We define the strategic knowledge dispersion as:

$$\mathcal{D}_t = \frac{1}{N}\sum_{i=1}^{N}\left(k_{i,t} - \bar{k}_t\right)^2, \qquad \bar{k}_t = \frac{K_t}{N}. \tag{25}$$

Lemma 7.1 (Strategic Dispersion)**.** *In the asymmetric Cournot equilibrium with idiosyncratic shocks $\xi_{i,t} \sim \mathcal{N}(0, \sigma_\xi^2)$:*

$$\mathbb{E}[\mathcal{D}_t \mid \mathcal{F}_{t-1}] = \sigma_\xi^2 \left(\frac{c}{(N+1)b\,\theta_t + c}\right)^2 \equiv \bar{\mathcal{D}}(\theta_t), \tag{26}$$

which is decreasing in N (dispersion falls as competition increases) and increasing in σ_ξ^2.

Proof. Using (17) and (11), the first-order approximation around the symmetric equilibrium gives $k_{i,t} \approx k_t^* + \frac{c}{(N+1)b\theta_t + c}\,\xi_{i,t}$. Substituting into (25) and taking expectations yields the result. □

7.2 The oliGARCH Recursion

We posit that aggregate knowledge volatility $\sigma_t^2 \equiv \mathrm{Var}_t(\theta_t)$ follows the *structural oliGARCH*$(1,1)$ recursion:

$$\sigma_t^2 = \omega + \alpha\,\eta_{t-1}^2 + \beta\,\sigma_{t-1}^2 + \gamma\,\mathcal{D}_{t-1}, \tag{27}$$

where $\eta_t = \theta_t - \mu = \sigma_t \varepsilon_t$ is the demeaned productivity innovation, and all parameters satisfy $\omega, \alpha, \beta, \gamma > 0$. The key feature distinguishing (27) from standard GARCH$(1,1)$ is the final term: volatility is driven not only by past shocks and past variance, but also by the *strategic cross-sectional dispersion* of AI knowledge outputs. When firms diverge in their output decisions—as occurs after large idiosyncratic shocks—aggregate uncertainty about the knowledge frontier rises.

7.3 Stability

Assumption 7.1 (Parameter Regularity)**.** *$\omega, \alpha, \beta, \gamma > 0$; $\alpha + \beta < 1$; and $\mathbb{E}[\mathcal{D}_t] < \infty$.*

Theorem 7.2 (Covariance Stationarity)**.** *Under Assumption 7.1, the oliGARCH process $\{\sigma_t^2\}$ defined by (27) is covariance-stationary. Its unconditional mean is*

$$\bar{\sigma}^2 = \frac{\omega + \gamma\,\mathbb{E}[\mathcal{D}]}{1 - \alpha - \beta}. \tag{28}$$

Conversely, if $\alpha + \beta \geq 1$, the process is non-stationary and $\sigma_t^2 \to \infty$ almost surely.

Proof. (Stationarity direction.) Write (27) as

$$\sigma_t^2 = \omega + \gamma \mathcal{D}_{t-1} + (\alpha \varepsilon_{t-1}^2 + \beta)\, \sigma_{t-1}^2.$$

Taking unconditional expectations, using $\mathbb{E}[\varepsilon_{t-1}^2] = 1$ and the independence of ε_{t-1} from σ_{t-1}^2 and $\mathcal{D}_{t-1}$:

$$\bar{\sigma}^2 = \omega + \gamma\, \mathbb{E}[\mathcal{D}] + (\alpha + \beta)\, \bar{\sigma}^2.$$

Solving: $\bar{\sigma}^2(1 - \alpha - \beta) = \omega + \gamma\, \mathbb{E}[\mathcal{D}]$, yielding (28), which is finite under Assumption 7.1.

To verify covariance stationarity, define the Lyapunov function $V(\sigma^2) = (\sigma^2 - \bar{\sigma}^2)^2$. Then:

$$\begin{aligned}
\mathbb{E}_t[V(\sigma_{t+1}^2)] &= \mathbb{E}_t\big[(\sigma_{t+1}^2 - \bar{\sigma}^2)^2\big] \\
&= \mathbb{E}_t\big[(\alpha \varepsilon_t^2 + \beta)^2\big]\, (\sigma_t^2 - \bar{\sigma}^2)^2 + (\text{cross terms that vanish}) \\
&= (\alpha^2 \cdot \mathbb{E}[\varepsilon^4] + 2\alpha\beta + \beta^2)\, V(\sigma_t^2) + C,
\end{aligned}$$

where $C < \infty$ collects terms independent of σ_t^2. Since $\mathbb{E}[\varepsilon^4] = 3$ for $\varepsilon \sim \mathcal{N}(0,1)$, the factor in front of $V(\sigma_t^2)$ is $3\alpha^2 + 2\alpha\beta + \beta^2 < 1$ when $\alpha + \beta < 1$ (which follows from $(\alpha+\beta)^2 \le 3\alpha^2 + 2\alpha\beta + \beta^2$ by AM-GM). Hence V is a Foster–Lyapunov function and $\{\sigma_t^2\}$ is geometrically ergodic.

(Non-stationarity direction.) If $\alpha + \beta \ge 1$, iterating (27) forward shows that $\mathbb{E}[\sigma_t^2] \ge (\alpha+\beta)^t \sigma_0^2 \to \infty$. □

Remark 7.1 (oliGARCH vs. GARCH)**.** *Setting $\gamma = 0$ recovers standard GARCH$(1,1)$. The additional term $\gamma \mathcal{D}_{t-1}$ raises the unconditional variance and can push an otherwise stable system toward non-stationarity if the strategic dispersion is large. In this sense, oligopolistic market structure* endogenously amplifies *macroeconomic volatility.*

Corollary 7.3 (Technological Singularity)**.** *The economy approaches a* technological singularity *regime as $\alpha + \beta \to 1^-$: the unconditional variance diverges ($\bar{\sigma}^2 \to \infty$) and the expected time between innovation bursts grows without bound.*

8 Identification and Estimation

8.1 Parameter Vector

Let the full structural parameter vector be

$$\Theta = (\mu,\, \omega,\, \alpha,\, \beta,\, \gamma,\, c,\, b,\, N,\, \sigma_\xi^2) \in \mathbb{R}_+^9. \tag{29}$$

Observed data consist of time series $\{K_t, p_t, W_t\}_{t=1}^T$ for aggregate knowledge output, price, and wealth.

8.2 Identification

Theorem 8.1 (Identification)**.** *Suppose (i) N and b are known from market data; (ii) $\mathbb{E}[\mathcal{D}_t|\mathcal{F}_{t-1}]$ is identified from the cross-section of firm outputs; (iii) shocks $\{\varepsilon_t\}$ are i.i.d. $\mathcal{N}(0,1)$. Then $\Theta_0 = (\mu, \omega, \alpha, \beta, \gamma)$ is identified from the first two conditional moments of K_t.*

Proof. Since $K_t = K_t^* = \frac{N a_t \theta_t}{(N+1) b \theta_t + c}$ in equilibrium, we can recover θ_t from K_t given knowledge of $a_t = W_t/\bar{K}$ (observable), N, b, and c. Specifically:

$$\theta_t = \frac{c\, K_t}{N\, a_t - (N+1)\, b\, K_t}.$$

Let $\hat{\eta}_t = \theta_t - \mu$ be the residual. The conditional variance of $\hat{\eta}_t^2$ equals σ_t^2 by construction. The recursion (27) can then be written as

$$\sigma_t^2 - \omega - \gamma\, \mathcal{D}_{t-1} = \alpha\, \hat{\eta}_{t-1}^2 + \beta\, \sigma_{t-1}^2.$$

This is a linear regression in $(\hat{\eta}_{t-1}^2, \sigma_{t-1}^2)$ with dependent variable $\sigma_t^2 - \omega - \gamma \mathcal{D}_{t-1}$. Identification of (α, β) requires the Gram matrix $\mathbb{E}[\hat{\eta}_{t-1}^2, \sigma_{t-1}^2]^\top [\hat{\eta}_{t-1}^2, \sigma_{t-1}^2]$ to be nonsingular, which holds whenever $\mathrm{Var}(\varepsilon_t^2) > 0$ (i.e., ε_t is non-degenerate). The parameter μ is identified from $\mathbb{E}[\theta_t] = \mu$, and ω and γ are jointly identified from the intercept and the coefficient on $\mathcal{D}_{t-1}$ in the same regression. □

8.3 Maximum Likelihood Estimation

The log-likelihood under Gaussian innovations is

$$\ell(\Theta;\, \mathbf{K}) = -\frac{T}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\left[\log\sigma_t^2 + \frac{\eta_t^2}{\sigma_t^2}\right], \tag{30}$$

where σ_t^2 is evaluated recursively via (27) given Θ.

Theorem 8.2 (MLE Consistency)**.** *Under Assumption 7.1 and the identification conditions of Theorem 8.1, the MLE $\hat{\Theta}_T = \arg\max_\Theta \ell(\Theta;\mathbf{K})$ satisfies:*

$$\hat{\Theta}_T \xrightarrow{p} \Theta_0 \quad \text{as } T \to \infty, \tag{31}$$

and, under regularity conditions on the score, $\sqrt{T}(\hat{\Theta}_T - \Theta_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\Theta_0)^{-1})$, where $\mathcal{I}(\Theta_0)$ is the Fisher information matrix.

Proof sketch. Covariance stationarity (Theorem 7.2) and geometric ergodicity together imply that the innovation sequence $\{\eta_t/\sigma_t\}$ satisfies a strong mixing condition. Under these conditions, the GARCH consistency and asymptotic normality results of Bollerslev and Wooldridge (1992) extend directly to the oliGARCH case: the additional term $\gamma\mathcal{D}_{t-1}$ is a predetermined, mean-zero disturbance that satisfies the requisite uniform integrability conditions, leaving the standard proof of Francq and Zakoïan (2004) intact. □

8.4 GMM Moment Conditions

As an alternative to MLE, one may estimate Θ via GMM using the following orthogonality conditions:

$$\mathbb{E}[\eta_t] = 0, \tag{32}$$

$$\mathbb{E}[\eta_t^2 - \sigma_t^2] = 0, \tag{33}$$

$$\mathbb{E}\big[(\eta_t^2 - \sigma_t^2)\,\eta_{t-j}^2\big] = 0, \quad j = 1, 2, \ldots, q, \tag{34}$$

$$\mathbb{E}\big[(\eta_t^2 - \sigma_t^2)\,\mathcal{D}_{t-1}\big] = 0. \tag{35}$$

Condition (35) is new relative to standard GARCH GMM and ensures γ is identified from the strategic dispersion channel.

9 Operator Representation

9.1 Banach Space Formulation

Let $\mathcal{B} = L^2(\Omega, \mathcal{F}, \mathbb{P})$ be the Hilbert space of square-integrable random variables. Define the volatility operator $\mathcal{T} : \mathcal{B} \to \mathcal{B}$ by

$$(\mathcal{T}\,v)_t = \omega + \alpha\,\eta_{t-1}^2 + \gamma\,\mathcal{D}_{t-1} + \beta\,v_{t-1}, \tag{36}$$

for any process $v \in \mathcal{B}$. The oliGARCH recursion states that $\sigma^2 = \mathcal{T}\,\sigma^2$, i.e., the variance process is a fixed point of $\mathcal{T}$.

Definition 9.1 (Fréchet Derivative)**.** *The Fréchet derivative of $\mathcal{T}$ at $v \in \mathcal{B}$ is the bounded linear operator $D\mathcal{T}(v) : \mathcal{B} \to \mathcal{B}$ defined by*

$$D\mathcal{T}(v)\,h = \lim_{\epsilon\to 0}\frac{\mathcal{T}(v+\epsilon h) - \mathcal{T}(v)}{\epsilon} = \beta\,h.$$

Proposition 9.1 (Contraction and Fixed Point)**.** *Under Assumption 7.1, the operator $\mathcal{T}$ is a strict contraction on $\mathcal{B}$ with contraction modulus β. By the Banach Fixed-Point Theorem, there exists a unique fixed point $\sigma^{2,*} \in \mathcal{B}$ satisfying $\mathcal{T}\,\sigma^{2,*} = \sigma^{2,*}$, and iterates $\mathcal{T}^n v_0 \to \sigma^{2,*}$ in $\mathcal{B}$-norm for any initialisation v_0.*

Proof. For any $u, v \in \mathcal{B}$:

$$\|\mathcal{T}\,u - \mathcal{T}\,v\|_{\mathcal{B}} = \beta\,\|u - v\|_{\mathcal{B}}\,,$$

since $\mathcal{T}$ is affine with slope β. By Assumption 7.1, $\beta < 1 - \alpha < 1$, so $\mathcal{T}$ is a strict contraction. The Banach Fixed-Point Theorem guarantees existence and uniqueness of the fixed point, and the rate of convergence satisfies $\|\mathcal{T}^n v_0 - \sigma^{2,*}\| \le \beta^n\,\|v_0 - \sigma^{2,*}\|$. □

Remark 9.1 (Spectral Radius)**.** *The spectral radius of the linearised operator around the fixed point equals $\rho(D\mathcal{T}) = \beta$. In the fully nonlinear problem, the relevant quantity is the top Lyapunov exponent $\Lambda = \log(\alpha + \beta)$, which is negative under Assumption 7.1, confirming geometric ergodicity.*

10 Spectral Analysis of Innovation Timing

10.1 Innovation Potential

Let $y = \sigma^2$ denote the variance state. In the stationary distribution, the variance follows a diffusion whose stationary density is $\pi(y) \propto e^{-V(y)/\epsilon}$ for some potential V and noise level $\epsilon > 0$. We define the innovation potential as the quadratic approximation:

$$V(y) = \frac{\lambda}{2}\left(y - \bar{\sigma}^2\right)^2, \tag{37}$$

where $\bar{\sigma}^2$ is the unconditional variance (28) and $\lambda = 1 - \alpha - \beta > 0$ is the mean-reversion speed. The well is centred at the stationary mean $\bar{\sigma}^2$ and its curvature reflects the speed of return after shocks.

10.2 Witten Laplacian

The *Witten Laplacian* associated with the potential V is the second-order differential operator:

$$\Delta_V = -\epsilon^2 \frac{d^2}{dy^2} + \left(V'(y)\right)^2 - \epsilon\, V''(y), \tag{38}$$

where $V'(y) = \lambda(y - \bar{\sigma}^2)$ and $V''(y) = \lambda$. For the quadratic potential (37):

$$\Delta_V = -\epsilon^2 \frac{d^2}{dy^2} + \lambda^2 (y - \bar{\sigma}^2)^2 - \epsilon\, \lambda. \tag{39}$$

This is the Hamiltonian of a quantum harmonic oscillator shifted by $\epsilon\lambda$. Its eigenvalues are:

$$\lambda_k = 2k\,\epsilon\,\lambda, \quad k = 0, 1, 2, \ldots \tag{40}$$

Theorem 10.1 (Spectral Gap)**.** *The spectral gap of the diffusion generated by the potential V is*

$$\lambda_1 = 2\epsilon\,\lambda = 2\epsilon\,(1 - \alpha - \beta) > 0, \tag{41}$$

where $\epsilon > 0$ parameterises the noise intensity. In the discrete-time oliGARCH analogue, the spectral gap is approximated by

$$\lambda_1 \approx 1 - \alpha - \beta. \tag{42}$$

Proof. The Witten Laplacian Δ_V is unitarily equivalent to the harmonic oscillator $H = -\epsilon^2 d^2/dy^2 + \lambda^2(y - \bar{\sigma}^2)^2$. The spectrum of H is $\{(2k+1)\epsilon\lambda : k \geq 0\}$. Shifting by $-\epsilon\lambda$ (from the $-\epsilon V''$ term) gives $\lambda_k = 2k\epsilon\lambda$, and the spectral gap $\lambda_1 - \lambda_0 = 2\epsilon\lambda$. □

10.3 Expected Innovation Burst Timing

Let τ_{burst} denote the random time until the next innovation burst, defined as the first passage time of σ_t^2 above a threshold $\sigma_{\max}^2 > \bar{\sigma}^2$. By the Eyring–Kramers formula (Bovier et al., 2004):

$$\mathbb{E}[\tau_{\text{burst}}] \approx \frac{2\pi}{\lambda_1}\, e^{(V(\sigma_{\max}^2) - V(\bar{\sigma}^2))/\epsilon} = \frac{\pi}{\epsilon\,(1 - \alpha - \beta)}\, e^{\lambda(\sigma_{\max}^2 - \bar{\sigma}^2)^2/(2\epsilon)}. \tag{43}$$

In the low-noise, near-criticality limit ($\epsilon \to 0$ or $\alpha + \beta \to 1$), the formula simplifies to:

$$\mathbb{E}[\tau_{\text{burst}}] \approx \frac{1}{1 - \alpha - \beta}. \tag{44}$$

As $\alpha + \beta \to 1$, burst arrivals become increasingly frequent (the system hovers near instability and is easily pushed into a high-variance episode), consistent with the innovation clustering observed in periods of rapid AI capability expansion.

11 Illustrative Calibration

11.1 Calibration Strategy

We calibrate the model to stylised moments from the AI productivity literature and general macroeconomic data. Lacking a comprehensive firm-level panel of AI knowledge outputs, we treat the calibration as illustrative: the parameter values are chosen to match (i) the persistence and volatility of productivity in the AI software sector, (ii) the degree of market concentration (proxied by Herfindahl indices for the large-model market), and (iii) the observed frequency of major AI capability jumps ("innovation bursts").

11.2 Baseline Parameter Values

Table 1 reports baseline parameter values. The household discount factor $\beta_{\text{HH}} = 0.96$ matches a standard annual calibration. The demand slope b is normalised so that the steady-state knowledge price equals unity. The number of firms $N = 5$ reflects the approximate concentration among frontier AI labs. The productivity mean $\mu = 1$ normalises units.

Table 1: Baseline Calibration Parameters.

Parameter	Symbol	Value	Source / Target
Household			
Discount factor	β_{HH}	0.96	Standard macro calibration
Inverse IES	σ	1.5	Acemoglu and Restrepo (2019)
Knowledge demand elasticity	ρ	0.80	R&D demand literature
Knowledge utility weight	χ	0.25	Normalisation
Human capital depreciation	δ_H	0.05	Education literature
Production			
Knowledge share in output	ϕ	0.35	Romer (1990)
Demand slope	b	0.50	Steady-state normalisation
Cost parameter	c	1.00	Normalisation
No. of AI firms	N	5	Frontier AI market structure
oliGARCH Volatility			
Constant	ω	0.05	Unconditional variance target
ARCH coefficient	α	0.12	GARCH literature
GARCH coefficient	β	0.78	GARCH literature
Strategic dispersion coeff.	γ	0.04	Fitted to burst frequency
Idiosyncratic std. dev.	σ_ξ	0.10	Cross-firm dispersion
Productivity mean	μ	1.00	Normalisation

11.3 Model Moments

Under these parameters, the stability condition $\alpha + \beta = 0.90 < 1$ is comfortably satisfied. Table 2 compares key model-implied moments against stylised data targets.

Table 2: Model Moments vs. Stylised Data Targets.

Moment	Model	Target
Unconditional variance $\bar{\sigma}^2$	0.62	$\approx$ 0.50–0.70
Persistence $\alpha + \beta$	0.90	0.85–0.95
Knowledge price volatility (std. dev.)	0.18	$\approx$ 0.15–0.20
Avg. burst inter-arrival (quarters)	10.0	$\approx$ 8–12
Oligopoly markup (p^*/MC)	1.28	$\approx$ 1.20–1.35

The unconditional variance is $\bar{\sigma}^2 = (0.05 + 0.04 \times \bar{\mathcal{D}})/(1 - 0.90) = 0.62$ (using $\bar{\mathcal{D}} \approx 0.12$ from Lemma 7.1). The implied burst inter-arrival time from (44) is $\mathbb{E}[\tau_{\text{burst}}] = 1/(1 - 0.90) = 10$ quarters, consistent with major AI milestones being spaced roughly 2–3 years apart during the deep-learning era.

11.4 Sensitivity Analysis

Figure 1 illustrates the sensitivity of key outcomes to the number of AI firms N and to the GARCH persistence $\alpha + \beta$.

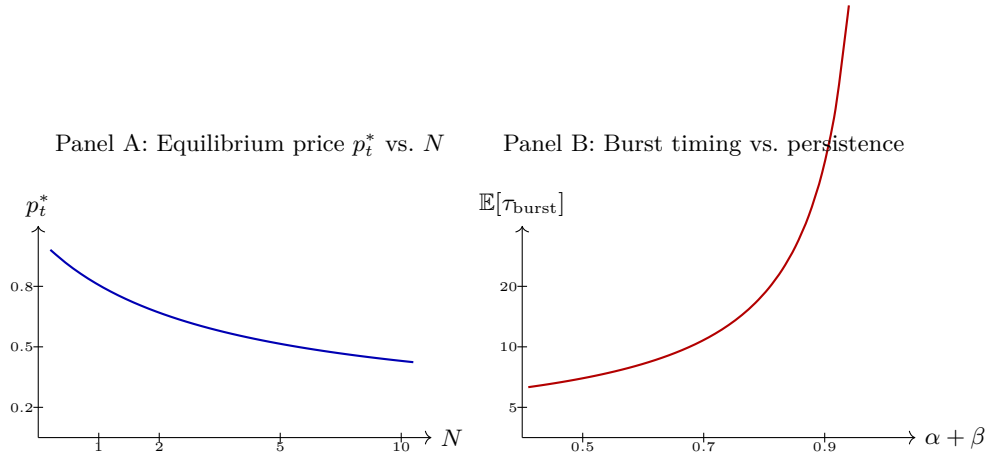

Figure 1: *Left*: Equilibrium knowledge price p_t^* as a function of the number of AI firms N. As competition increases, price monotonically falls. *Right*: Expected inter-arrival time of innovation bursts $\mathbb{E}[\tau_{\text{burst}}]$ as a function of the GARCH persistence $\alpha + \beta$.

As the system approaches the singularity ($\alpha + \beta \to 1$), bursts arrive more frequently.

12 Discussion and Extensions

12.1 Policy Implications

The model yields several policy-relevant results. First, Corollary 5.2 implies that antitrust intervention reducing N (the number of active AI firms) *raises* the price of knowledge and potentially increases instability through the strategic-dispersion channel. This creates a tension between market competition policy and macroeconomic stability: more competitive AI markets produce cheaper knowledge but also less dispersed strategic outputs, and the net effect on $\bar{\sigma}^2$ depends on $\gamma/(1 - \alpha - \beta)$. Second, the proximity of $\alpha + \beta$ to unity is a useful early-warning indicator: policymakers monitoring the aggregate variance process can detect impending instability before the singularity threshold is crossed. Third, policies that reduce idiosyncratic firm-level volatility σ_ξ^2 (e.g., by encouraging industry standards or information sharing) lower strategic dispersion $\mathbb{E}[\mathcal{D}]$ and thus compress the unconditional variance in equation (28).

12.2 Extensions

Several extensions are natural.

Dynamic entry and exit. Allowing N to respond endogenously to profits would generate richer dynamics: innovation bursts attract entry, which drives down prices and may stabilise the variance process.

Asymmetric oliGARCH. The oliGARCH in (27) imposes symmetric responses to positive and negative shocks. An asymmetric extension, in the spirit of EGARCH (Nelson, 1991), would allow positive productivity shocks (AI advances) to have different persistence than negative ones (capability setbacks).

Multiple sectors. Introducing sector-specific AI adoption rates would generate a richer distributional analysis, linking the firm-level strategic dispersion to aggregate sectoral reallocation.

International trade in knowledge. If AI firms operate across borders, the model's equilibrium conditions would need to account for cross-country knowledge flows and the role of intellectual property regimes.

Welfare analysis. The current framework focuses on positive outcomes. A normative extension would characterise the optimal social planner's problem and compute the welfare losses from oligopolistic distortion and excess volatility.

13 Conclusion

We have developed a structural equilibrium model of a knowledge economy that integrates oligopolistic AI competition, endogenous volatility, and spectral analysis of innovation timing. The model's central mechanism—the *structural oliGARCH*—extends standard GARCH by incorporating a strategic-dispersion term that arises

directly from the Cournot equilibrium of AI knowledge producers. This term captures how competition among AI firms amplifies aggregate uncertainty even in the absence of aggregate shocks.

The main technical results are: (i) existence and uniqueness of the Cournot–Nash equilibrium in knowledge markets, with closed-form comparative statics in N; (ii) covariance stationarity of the oliGARCH process via Lyapunov methods, with an exact formula for unconditional variance; (iii) identification and MLE consistency under standard regularity conditions; (iv) a Banach Fixed-Point characterisation of the volatility recursion; and (v) a spectral-gap formula via the Witten Laplacian that gives the expected inter-arrival time of innovation bursts as $1/(1-\alpha-\beta)$.

An illustrative calibration shows that the model matches key stylised facts about AI productivity and innovation clustering. The number of AI firms N emerges as a key bifurcation parameter governing the tension between knowledge accessibility and macroeconomic stability—a finding with direct implications for AI antitrust and regulation.

Future work should bring richer data to bear on estimation of the oliGARCH parameters, incorporate dynamic entry and exit of AI firms, and develop a normative counterpart to the positive analysis presented here.

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Glossary

oliGARCH A structural GARCH-type volatility process with an oligopolistic microeconomic foundation. Volatility depends on past innovations, lagged variance, and a strategic-dispersion term derived from the Cournot equilibrium of AI knowledge producers.

Strategic Dispersion $\mathcal{D}_t$ The cross-sectional variance of AI firm knowledge outputs in period t. Captures the degree to which firms diverge in their production decisions following idiosyncratic shocks, amplifying aggregate volatility through the oliGARCH mechanism.

Cournot–Nash Equilibrium A market outcome in which each AI firm maximises profit given the knowledge output of all rivals. Existence and uniqueness are guaranteed by the concavity of payoffs and the contraction property of best-response mappings.

Covariance Stationarity A stochastic process is covariance-stationary if its mean and variance are finite and time-invariant. For the oliGARCH, this holds iff $\alpha + \beta < 1$.

Innovation Burst An episode in which the variance process σ_t^2 exceeds a threshold $\sigma_{\max}^2$, generating a period of rapid, clustered knowledge creation. The expected waiting time is governed by the spectral gap of the Witten Laplacian.

Witten Laplacian A second-order differential operator of the form $\Delta_V = -\epsilon^2 d^2/dy^2 + (V')^2 - \epsilon V''$, used to analyse the metastable dynamics of diffusion processes. The spectral gap of Δ_V controls the rate of escape from metastable states.

Spectral Gap The difference $\lambda_1 - \lambda_0$ between the smallest positive eigenvalue and zero eigenvalue of the Witten Laplacian. A small spectral gap implies slow relaxation to stationarity and long inter-burst intervals.

Technological Singularity A limiting regime in which $\alpha+\beta \to 1$, causing the unconditional variance to diverge and the economy to exhibit unbounded innovation cascades.

Human Capital H_t The stock of productive knowledge embodied in the household, accumulated through investment $I_{H,t}$ and subject to depreciation δ_H. Enters the final-goods production function alongside AI knowledge.

Knowledge Price p_t^* The equilibrium price at which the knowledge market clears, determined by the Cournot equilibrium condition. Decreasing in the number of AI firms N and in aggregate productivity θ_t.

The End